BAHUM107 Macro Economics

Module 2 Graph Explanations

A detailed companion for VIT CAT and FAT preparation

This guide explains the graphs used to study income, employment and inflation. Read each page in order: identify the axes, understand the curves, locate the equilibrium, then trace the change and state its economic meaning. All diagrams and numerical examples are illustrative.

Graph guide

The numbers below refer to pages in this guide.

02 Classical Labour Market Equilibrium

03 Classical Production Function

04 Classical Output and the Price Level

05 Say's Law and Saving Investment Equilibrium

06 Pigou and Wage Price Flexibility

07 Keynesian Effective Demand

08 Keynesian Expenditure Output Equilibrium

09 Keynesian Wage Price Rigidity

10 Demand Pull Inflation

11 Cost Push Inflation

12 Inflation and the Purchasing Power of Money

13 Demand Driven Deflation

14 Stagflation and the Supply Shock Dilemma

15 Short Run Phillips Curve

16 Long Run Phillips Curve and Expectations

17 Comparing the graphs and avoiding common errors

18 Drawing method and source guide

The syllabus names topics rather than an exhaustive list of figures. This guide covers the graphs in your original Module 2 notes and adds three supporting diagrams: the production function, classical long-run AD–AS, and the Keynesian expenditure cross.

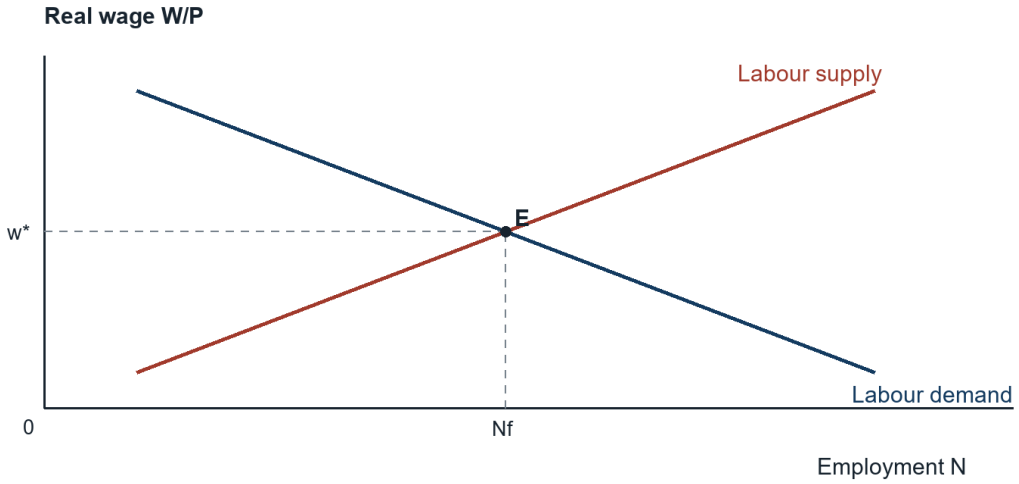
For exam answers, use the relevant graph and a focused explanation. The fuller explanations here are for understanding and revision; they are not intended to be copied in full into every 10-mark answer.

Graph 01 Classical Labour Market Equilibrium

How the classical model determines the real wage and full-employment labour.

Axes and symbols. The horizontal axis measures employment N; the vertical axis measures the real wage W/P. W is money wages and P is the price level. Labour demand represents employers' hiring plans; labour supply represents workers' willingness to work.

Why the curves have these shapes. Labour demand slopes downward because an additional worker normally adds less output when capital is fixed. A lower real wage makes hiring more workers profitable. Labour supply is drawn upward: in this simplified model, higher real wages encourage more labour to be offered.



Reading the equilibrium. E is the intersection of demand and supply. The horizontal dashed line identifies the equilibrium real wage w*. The vertical dashed line identifies employment Nf. At this wage, firms' planned hiring equals workers' planned labour supply.

Adjustment and interpretation. A real wage above w* creates excess labour supply: more people want jobs than firms want to hire. Classical wage competition pushes the wage downward. Below w*, firms want more labour than is offered, putting upward pressure on wages. These are movements along unchanged curves. Better technology or a change in available workers can instead shift a curve.

Illustration. If firms and workers both choose 100 workers at a real wage of 5 units of output per worker, the market clears. A higher real wage might reduce hiring while attracting more job seekers. These numbers illustrate the mechanism rather than calibrate the plotted curves.

Exam conclusion. Flexible real wages bring labour demand and labour supply into equality. Given capital and technology, this employment level determines classical full-employment output.

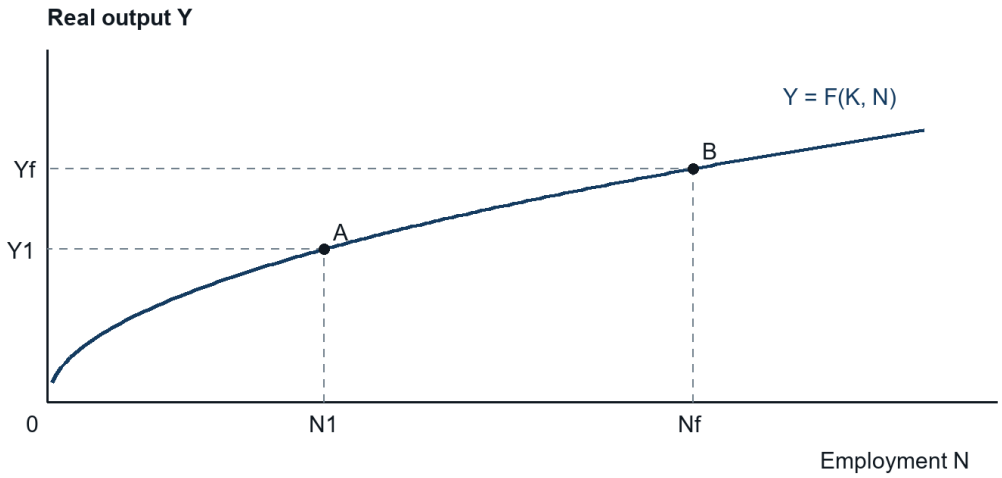
Common error. Do not put money wages on the vertical axis while explaining real-wage adjustment. Full employment also does not mean zero frictional or structural unemployment.

Graph 02 Classical Production Function

How employment is translated into real output once capital and technology are given.

Axes and symbols. Employment N is on the horizontal axis and real output Y is on the vertical axis. The label Y = F(K, N) means output depends on capital K and labour N. Capital and technology are held constant for this curve.

Why the curves have these shapes. The curve rises because employing more labour increases production. It becomes flatter because successive additions of labour contribute smaller increments of output when workers share a fixed capital stock. This is diminishing marginal productivity, not falling total output.



Reading the equilibrium. Take Nf from the labour-market graph. Move vertically from Nf until reaching the production curve, then horizontally to the output axis. This gives full-employment output Yf. The point is a production mapping, not an intersection of demand and supply curves.

Adjustment and interpretation. Moving from N1 to Nf raises output from Y1 to Yf along the same production function. A better technology or larger capital stock can shift the function upward, allowing more output from the same employment. A wage change alone does not shift this technological relationship.

Illustration. With fixed machines, adding workers might raise output from 80 to 100, then to 115 units: output is increasing, but the extra amounts are getting smaller. These are illustrative quantities. The plotted Nf identifies the labour level supplied by the preceding classical model.

Exam conclusion. The classical sequence is labour-market equilibrium, then employment Nf, then output Yf through the production function. Real income corresponds to real output in the simplified model.

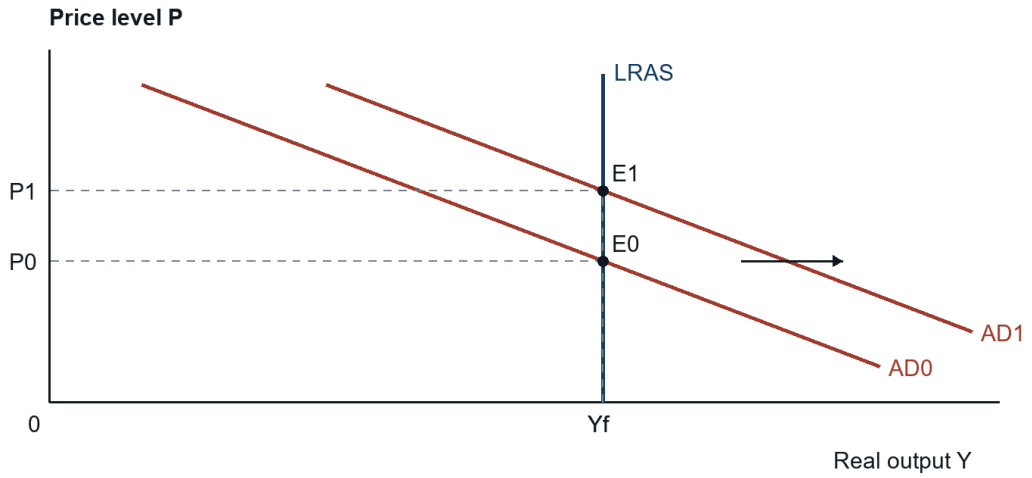
Common error. Do not call this an aggregate-supply curve: its horizontal variable is employment, not output, and its vertical variable is output, not the price level. Do not infer that diminishing marginal productivity means total output must decrease.

Graph 03 Classical Output and the Price Level

How a demand change affects prices when long-run real output is fixed by productive capacity.

Axes and symbols. The horizontal axis is real output Y and the vertical axis is the general price level P. LRAS is long-run aggregate supply at potential output Yf. AD0 and AD1 show aggregate demand before and after an increase in spending.

Why the curves have these shapes. LRAS is vertical because long-run productive capacity depends on labour, capital and technology, rather than the price level. AD slopes downward: a higher price level generally reduces the quantity of domestic output demanded through purchasing-power, interest-rate and external-demand channels.



Reading the equilibrium. The intersection of AD0 and LRAS is E0, with output Yf and price level P0. The intersection of AD1 and the same LRAS is E1. Both equilibrium points project to the same output on the horizontal axis.

Adjustment and interpretation. When aggregate demand shifts right, equilibrium moves upward from E0 to E1. The price level rises to P1, but long-run output remains Yf. A leftward demand shift would lower the equilibrium price level under the model's flexible-price assumptions. A genuine increase in productive capacity would shift LRAS right instead.

Illustration. Suppose sustainable output is 1,000 units and total spending expands. After full wage and price adjustment, spending cannot permanently make the same resources produce unlimited output; the new equilibrium has a higher price level. The graph abstracts from short-run output changes during adjustment.

Exam conclusion. In the classical long-run benchmark, aggregate demand determines the price level while productive resources determine real output. Demand expansion does not permanently raise output above potential.

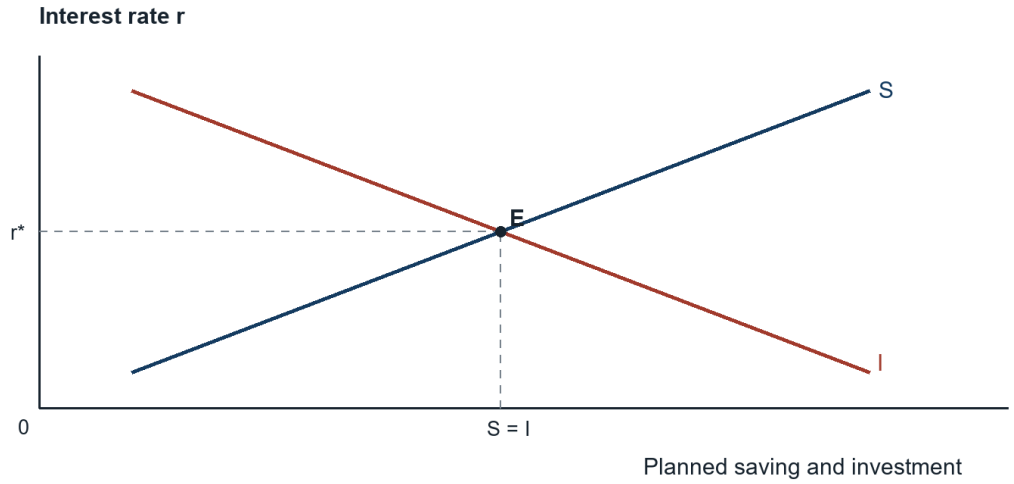
Common error. Do not apply this final long-run result to every short-run situation. A single shift compares price levels; it does not by itself establish a permanently positive inflation rate.

Graph 04 Say's Law and Saving Investment Equilibrium

How the classical interest-rate mechanism is used to reconcile planned saving with planned investment.

Axes and symbols. The horizontal axis measures planned saving and investment per period. The vertical axis measures the interest rate r. S is the saving schedule and I is the investment schedule. Both schedules refer to a given setting, including the assumed income level.

Why the curves have these shapes. S slopes upward in the conventional classical model because a higher interest rate encourages saving. I slopes downward because a higher borrowing cost makes fewer investment projects profitable. These are model assumptions; the saving response need not be identical for every household.



Reading the equilibrium. The schedules intersect at E. The horizontal projection gives r*, and the vertical projection gives the common amount S = I. At this interest rate, funds households plan to save match the investment businesses plan to undertake.

Adjustment and interpretation. Above r*, saving exceeds investment, creating downward pressure on the interest rate. As r falls, planned saving falls and investment rises along their schedules. Below r*, investment exceeds saving, putting upward pressure on r. If willingness to save increases at each rate, S shifts right and the new equilibrium interest rate tends to fall.

Illustration. At income of Rs 1,000 crore, households may consume Rs 800 crore and save Rs 200 crore. If firms invest Rs 200 crore, consumption plus investment buys the output. Say's Law relies on a mechanism that turns the saved income into spending.

Exam conclusion. Interest-rate flexibility is the classical explanation for why saving need not cause a lasting shortage of aggregate expenditure.

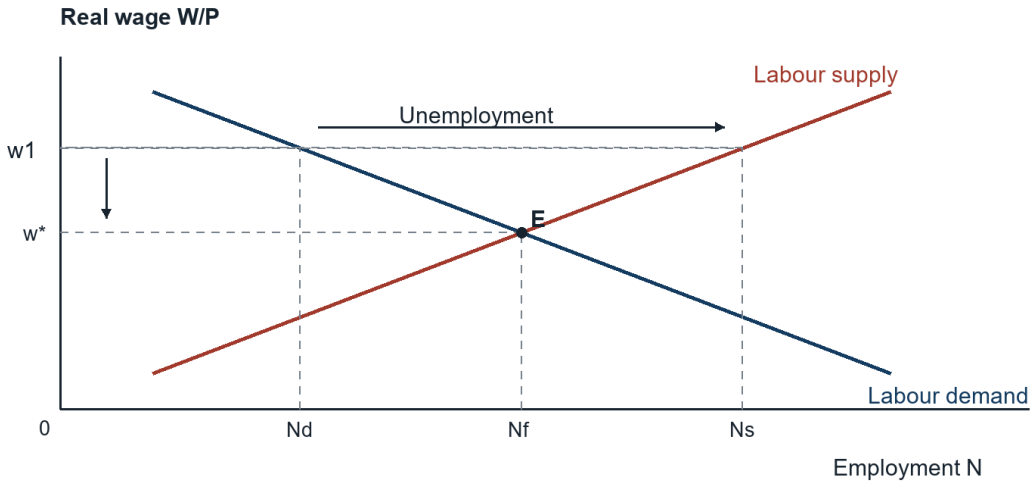
Common error. Do not use the accounting identity of actual saving and investment as proof that planned saving and investment are equal at full employment. Unplanned inventories can make actual investment differ from planned investment.

Graph 05 Pigou and Wage Price Flexibility

How an excessively high real wage creates excess labour supply and how classical wage adjustment removes it.

Axes and symbols. Employment is on the horizontal axis and the real wage W/P is on the vertical axis. w1 is a wage above equilibrium w*. Nd and Ns on the horizontal axis identify labour demanded and supplied at w1, rather than two equilibrium employment levels.

Why the curves have these shapes. Labour demand slopes down and labour supply slopes up under the classical assumptions. The horizontal line through w1 cuts demand on the left and supply on the right. Thus, at the same wage, firms wish to hire fewer workers than the number offering labour.



Reading the equilibrium. The distance Ns minus Nd represents excess labour supply at w1. E is the lower intersection of the two curves. At E the real wage is w*, labour demand equals supply, and employment is Nf.

Adjustment and interpretation. Follow the downward arrow from w1 towards w*. Firms move down the demand curve and increase hiring; workers' planned labour supply adjusts along the supply curve. The unemployment gap closes at E. This conclusion assumes the real wage actually falls and aggregate expenditure remains adequate.

Illustration. If money wages fall by 10 per cent while prices remain unchanged, real wages fall by 10 per cent. If wages and prices both fall by 10 per cent, W/P is unchanged. Therefore, a nominal wage cut need not produce the movement required by this graph.

Exam conclusion. Pigou's wage-flexibility argument attributes unemployment in this model to a real wage above equilibrium and proposes real-wage adjustment as the corrective mechanism.

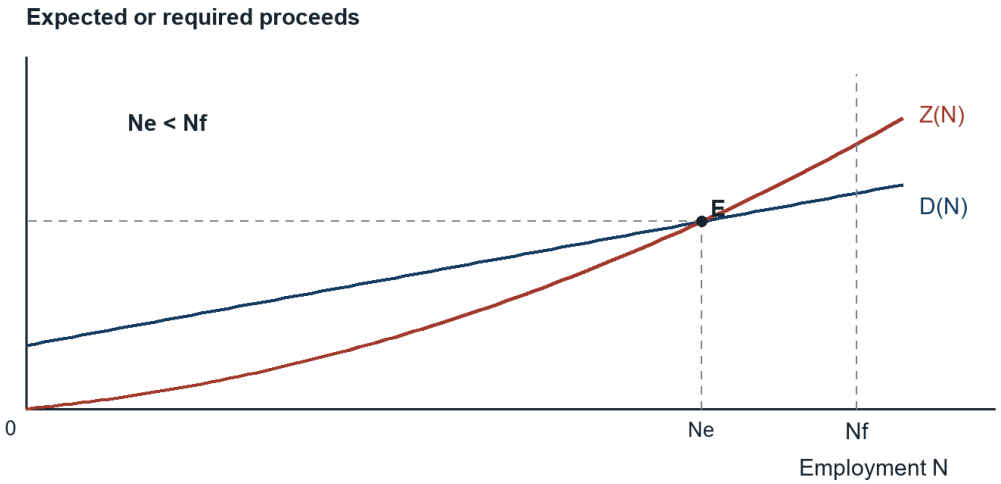
Common error. Do not confuse this with Pigou's real-balance effect, which concerns falling prices raising the purchasing power of money holdings. Keynes also warned that economy-wide wage cuts can reduce consumption and expected sales.

Graph 06 Keynesian Effective Demand

Why equilibrium employment can be below full employment when expected sales are insufficient.

Axes and symbols. Employment N is on the horizontal axis. The vertical axis measures total expected or required proceeds. D(N) is expected sales revenue from employing N workers. Z(N) is the minimum expected revenue that makes that employment worthwhile. These are revenue schedules, not price-index curves.

Why the curves have these shapes. D(N) rises as employment, income and expected expenditure rise. Z(N) also rises because employing more workers and producing more output requires greater proceeds to make production worthwhile. The schematic Z curve becomes steeper, so the schedules cross before full employment.



Reading the equilibrium. At E, D(N) equals Z(N). Dropping a vertical line gives equilibrium employment Ne. The separate vertical line marks full-employment Nf farther right. The distance between Ne and Nf represents the shortfall of equilibrium employment from its full-employment benchmark.

Adjustment and interpretation. Below Ne, D lies above Z: expected proceeds exceed the required proceeds, encouraging firms to expand employment. Above Ne, Z exceeds D, so firms reduce employment. These responses bring the economy towards E without necessarily bringing it to Nf. Higher planned expenditure can shift D upward and move the intersection right.

Illustration. An economy might support only 80 workers through expected sales even though its full-employment benchmark is 100. Hiring the extra workers is not profitable unless expected purchases support the additional production. The numbers are illustrative, not plotted measurements.

Exam conclusion. Effective demand determines equilibrium employment and therefore output and income. Because Ne can be below Nf, equilibrium does not guarantee full employment.

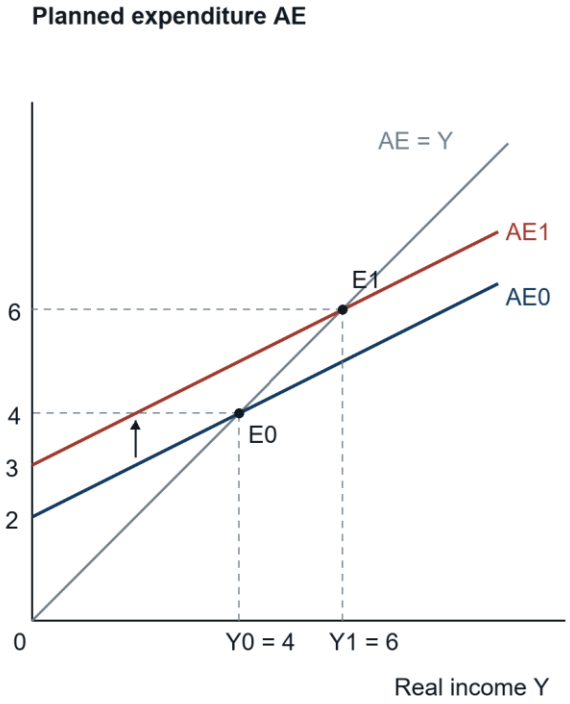
Common error. Do not label the vertical axis simply P. Also, do not assume wage stickiness is the entire argument: Keynes's central point is that effective demand can be deficient.

Graph 07 Keynesian Expenditure Output Equilibrium

How planned spending and inventory changes determine equilibrium income in the fixed-price model.

Axes and symbols. Real output or income Y is on the horizontal axis and planned expenditure AE on the vertical axis. The two axes use the same units and scale. The 45-degree line represents AE = Y, while AE0 and AE1 represent different planned spending schedules.

Why the curves have these shapes. The equality line rises one-for-one: every point has identical expenditure and output values. The AE schedules rise more slowly because only part of additional income is spent on consumption. They have positive intercepts because some planned spending does not depend on current income.



Reading the equilibrium. AE0 intersects the equality line at E0, giving Y0. At that point, firms' total output equals planned purchases. There is no unintended inventory change. With higher planned spending, AE1 intersects at E1 and supports a higher income Y1.

Adjustment and interpretation. To the right of E0 on the initial schedule, output exceeds planned purchases: inventories accumulate unexpectedly and firms cut production. To the left, planned purchases exceed current output: inventories fall and firms expand. A rise in autonomous spending shifts the AE schedule upward; adjustment then leads to E1.

Illustration. The illustration uses AE0 = 2 + 0.5Y and AE1 = 3 + 0.5Y. Initial equilibrium is Y0 = 4, because 2 + 0.5 × 4 = 4. After the upward shift, Y1 = 6, because 3 + 0.5 × 6 = 6. Units are arbitrary and prices are held fixed.

Exam conclusion. Income adjusts until planned expenditure equals production. The equilibrium can be below potential output; this diagram alone does not show where full employment lies.

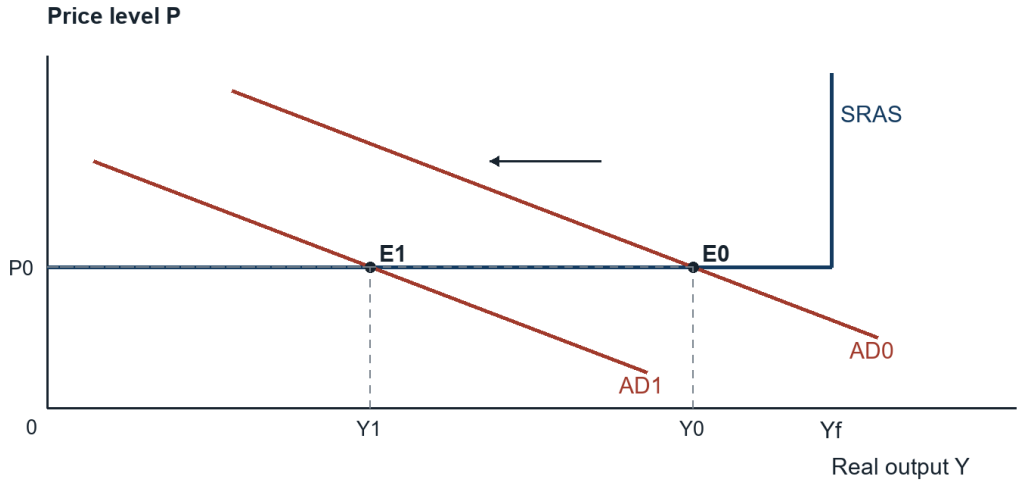
Common error. The 45-degree line is an expenditure-equality line, not the upward-sloping SRAS curve from a price-output graph. Multiplier derivations belong to the later module; here the purpose is equilibrium adjustment.

Graph 08 Keynesian Wage Price Rigidity

How weak demand can reduce output and employment while the price level remains sticky.

Axes and symbols. Real output Y is on the horizontal axis and the price level P on the vertical axis. SRAS has a horizontal segment at P0 and a vertical segment at potential output Yf. AD0 is initial demand; AD1 is lower demand.

Why the curves have these shapes. The horizontal segment is an extreme fixed-price Keynesian simplification: firms can expand or contract output using spare capacity without changing the price level. The vertical segment indicates the limit imposed by productive capacity. The demand curves slope downward in price-output space.



Reading the equilibrium. Initial demand intersects SRAS at E0, giving Y0 and P0. After demand weakens, AD1 intersects the same horizontal segment at E1. Output is now Y1, which is below Y0, but the price level remains P0.

Adjustment and interpretation. The leftward AD shift represents weaker planned spending, perhaps because businesses cancel investment. Firms experience lower sales and reduce production, hours and employment. With wages and prices slow to adjust, there is no immediate price reduction sufficient to restore sales and employment. A demand recovery reverses this movement while spare capacity remains.

Illustration. A factory with annual wage agreements may cancel shifts when orders fall. Its prices and wage contracts remain unchanged initially, so lower demand appears mainly as lower output. If money wages stay fixed while prices do fall, real wages rise, which can further discourage hiring.

Exam conclusion. When wages and prices are sticky, changes in aggregate demand may have large effects on output and employment. Market clearing through immediate nominal adjustment cannot be assumed.

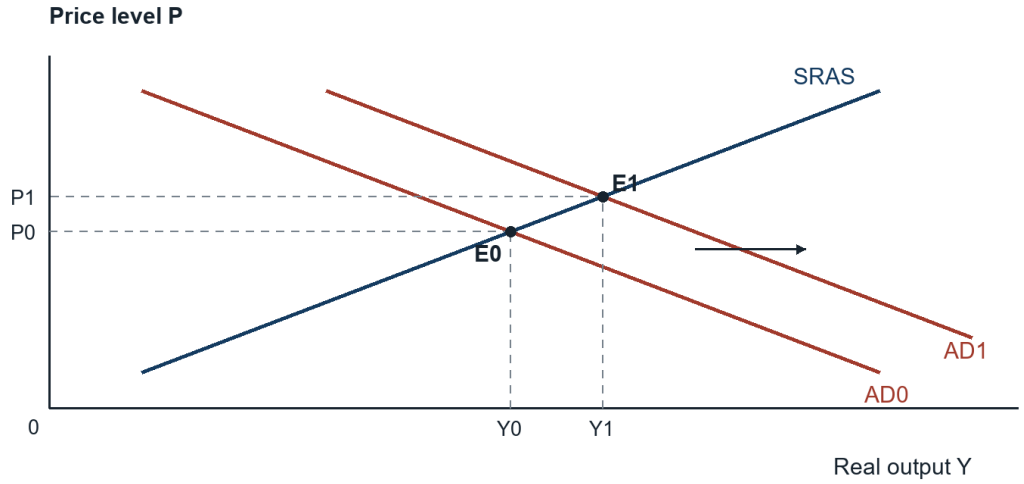
Common error. Do not claim every real-world price is permanently fixed. An upward-sloping SRAS is a more general short-run case. The horizontal-vertical shape is used to make the fixed-price mechanism clear.

Graph 09 Demand Pull Inflation

Why stronger aggregate demand raises the price level and can raise output in the short run.

Axes and symbols. The horizontal axis is real output Y and the vertical axis is the general price level P. AD0 and AD1 show demand before and after an expansion. SRAS is unchanged, so the diagram isolates a demand-side disturbance.

Why the curves have these shapes. SRAS slopes upward because supplying more output in the short run can require higher marginal costs, overtime or more intensive use of resources. AD slopes downward. A shift from AD0 to AD1 means greater planned spending at each price level, not simply a movement caused by a price change.



Reading the equilibrium. E0 gives the initial P0 and Y0. E1, where AD1 intersects SRAS, gives P1 and Y1. Comparing the dashed projections shows P1 above P0 and Y1 to the right of Y0. Output and the price level both rise.

Adjustment and interpretation. Higher consumption, investment, government purchases or export demand can shift AD right. Firms respond to more orders by expanding production and raising prices as costs and resource pressures rise. With ample spare capacity, output may respond strongly. Near capacity, the price response is generally greater. In the long run, unchanged productive capacity limits sustainable output.

Illustration. For example, a broad credit-supported spending expansion can increase orders across industries. If factories and workers are already heavily utilised, suppliers cannot easily meet every extra order, and prices rise. No numerical axis scale is needed to show these qualitative changes.

Exam conclusion. Demand-pull inflationary pressure begins with an outward demand shift. Its short-run signature is a higher price level accompanied by higher output, given an unchanged upward-sloping SRAS.

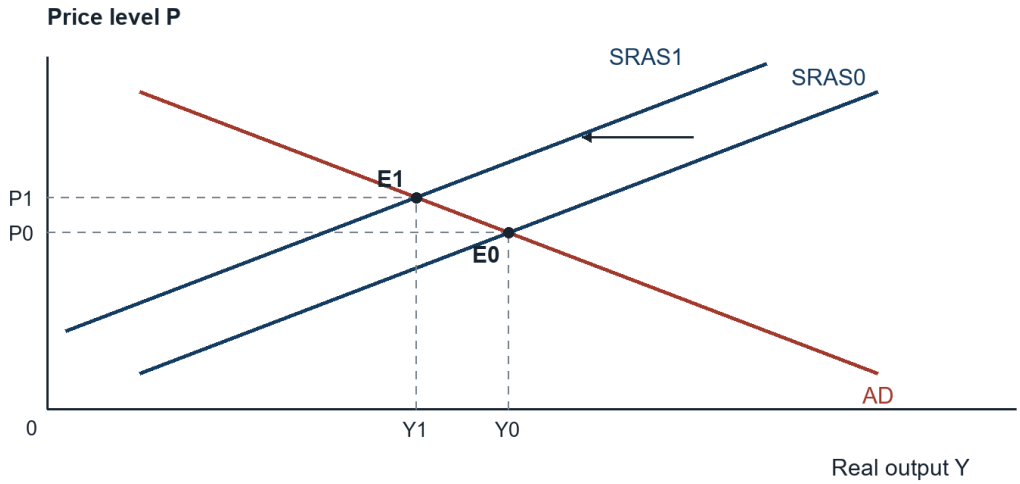
Common error. Do not shift SRAS to explain the initial demand shock. A single shift shows a price-level increase; continuing inflation requires ongoing demand pressure or other reinforcing developments.

Graph 10 Cost Push Inflation

Why an adverse supply shock raises prices while reducing output.

Axes and symbols. Real output Y is horizontal and the price level P is vertical. AD remains unchanged. SRAS0 is the initial supply curve and SRAS1 is the supply curve after costs rise or production becomes more difficult.

Why the curves have these shapes. Both SRAS curves slope upward, but SRAS1 lies to the left or above SRAS0. At a given price, firms supply less; equivalently, supplying a given output now requires a higher price. This is an entire curve shift, not a movement along SRAS0.



Reading the equilibrium. The initial intersection E0 gives P0 and Y0. The new intersection E1 gives P1 and Y1. The dashed lines show that P1 is higher and Y1 is lower. Employment generally falls as firms reduce production.

Adjustment and interpretation. Higher fuel costs, disrupted input supplies, crop failures or increased import costs can shift SRAS left. Firms pass part of the increased cost to buyers while cutting output. This differs from demand-pull pressure, where output initially rises. Wage increases cause this pressure only when they raise unit labour costs, for example by exceeding productivity growth.

Illustration. Suppose imported oil becomes more expensive. Transport, manufacturing and farming face higher costs. With aggregate spending initially unchanged, the economy produces less output at a higher price level. The graph represents the combined economy-wide effect rather than the price of oil alone.

Exam conclusion. An adverse supply shock produces a higher price level and lower real output. If inflation continues alongside weak activity and high unemployment, the result is stagflation.

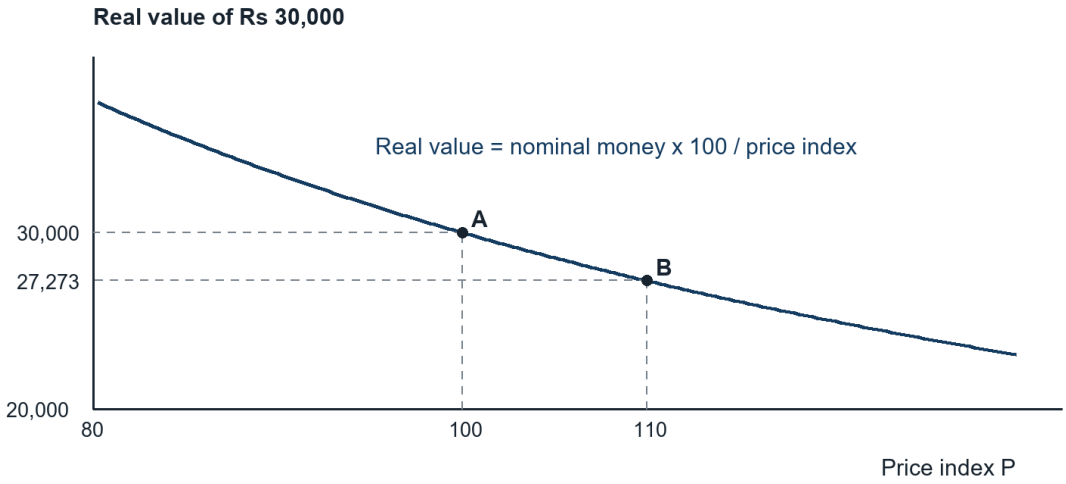
Common error. Do not assume that a one-time input-cost increase generates inflation forever. Persistent inflation needs repeated shocks or reinforcing mechanisms. Also distinguish a change in the general price level from an isolated product-price rise.

Graph 11 Inflation and the Purchasing Power of Money

How a higher price index reduces the real value of a fixed nominal amount.

Axes and symbols. The horizontal axis is the price index P, with the base period equal to 100. The vertical axis is the purchasing power of a fixed Rs 30,000, expressed in base-period rupees. The axes shown start at 80 and Rs 20,000, not at zero.

Why the curves have these shapes. The curve slopes downward because the same nominal money buys fewer goods when prices rise. The relationship is inverse: real value equals nominal money multiplied by 100 and divided by the price index. It is curved rather than a straight proportional decline.



Reading the equilibrium. A shows a price index of 100 and real value of Rs 30,000. B shows an index of 110 and real value of about Rs 27,273. These are points on a mathematical relationship, not market-clearing equilibria. There is no demand-supply intersection in this figure.

Adjustment and interpretation. A rise in P moves the holder of unchanged nominal money from A towards B along the curve. If nominal money itself changes, a different purchasing-power curve is needed. With both nominal income and prices rising by 10 per cent, purchasing power is preserved rather than reduced.

Illustration. At P = 110, calculate 30,000 × 100 ÷ 110 = Rs 27,272.73. The loss from Rs 30,000 is about 9.09 per cent. A 10 per cent price increase therefore does not mean an exactly 10 per cent decline in purchasing power.

Exam conclusion. Inflation erodes fixed nominal incomes and cash balances. Whether a particular household loses real income depends on how its nominal income changes relative to prices.

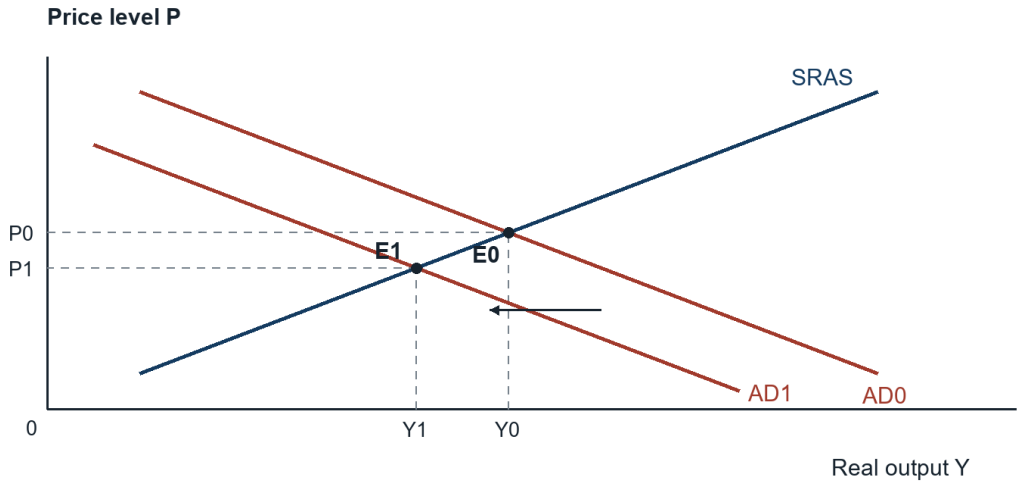
Common error. This figure illustrates one effect of inflation. It does not measure menu costs, uncertainty, or redistribution between debtors and creditors. Do not read the truncated axes as starting at zero.

Graph 12 Demand Driven Deflation

How a fall in aggregate demand can lower both output and the general price level.

Axes and symbols. The horizontal axis is real output Y and the vertical axis is the price level P. SRAS is held unchanged and upward sloping. AD0 is initial demand; AD1 is lower demand following a contraction in spending.

Why the curves have these shapes. AD1 lies left of AD0 because households, firms or other buyers plan to spend less at each price level. Moving to a lower price on AD0 would be a movement along that curve; it would not represent this independent fall in demand.



Reading the equilibrium. E0 identifies initial output Y0 and price level P0. After AD shifts left, E1 identifies lower output Y1 and lower price level P1. Both dashed projections move towards smaller values. The output decline normally weakens employment.

Adjustment and interpretation. Reduced confidence, investment or credit can initiate the shift. Falling sales lead firms to reduce production and prices. Expected further price declines can delay purchases; lower prices increase the real burden of fixed nominal debt. These channels can reinforce weak demand, although the single diagram shows only the initial change between two equilibria.

Illustration. For example, widespread attempts to repay debt may reduce purchases. If an index falls from 100 to 97, the measured price change is minus 3 per cent. This is deflation. Inflation slowing from 8 to 4 per cent would instead be disinflation, with prices still rising.

Exam conclusion. Demand-driven deflation is associated with falling prices, falling output and weaker employment. Its cause is deficient spending, represented by the leftward AD shift.

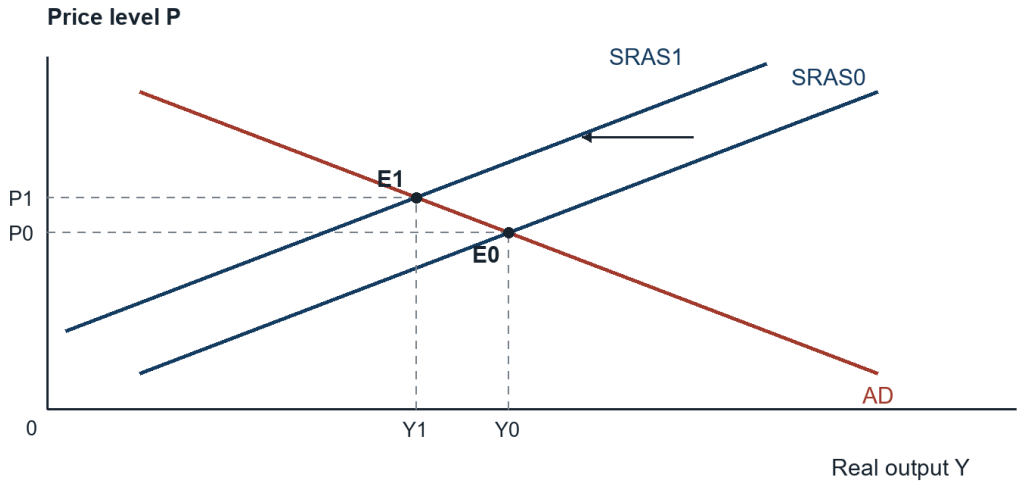
Common error. Do not claim all falling prices imply recession. A favourable supply shift can lower prices while raising output. Also, sustained deflation is a continuing fall in prices, not merely a low inflation rate.

Graph 13 Stagflation and the Supply Shock Dilemma

Why inflation and high unemployment can occur together despite weak economic activity.

Axes and symbols. The axes are the price level P and real output Y. SRAS0 shifts left to SRAS1 while AD is initially unchanged. The construction matches the cost-push graph because an adverse supply shock can cause both cost-push pressure and stagflation.

Why the curves have these shapes. The higher supply curve means firms need higher selling prices to cover increased costs at each output level. Demand has not independently increased. Thus, the new price-output combination is driven by worsening production conditions rather than a demand boom.



Reading the equilibrium. At E0 the economy has P0 and Y0. At E1 the price level is higher at P1 and output is lower at Y1. Lower production tends to reduce employment. When inflation persists together with stagnation and high unemployment, the economy is experiencing stagflation.

Adjustment and interpretation. Repeated input shocks or wage-price feedback can prolong the problem. Stimulating AD may support output but increase price pressure. Restricting AD may reduce price pressure but worsen the output loss. These possible responses explain the policy dilemma; no additional policy shift is drawn in the figure.

Illustration. A sustained disruption to fuel supplies can raise transport and production costs. Firms increase prices, reduce production and employ fewer workers. People therefore face rising living costs at the same time that jobs become harder to find.

Exam conclusion. Stagflation combines inflation with weak output and high unemployment. Its adverse-supply explanation differs from demand-pull inflation, where output initially increases.

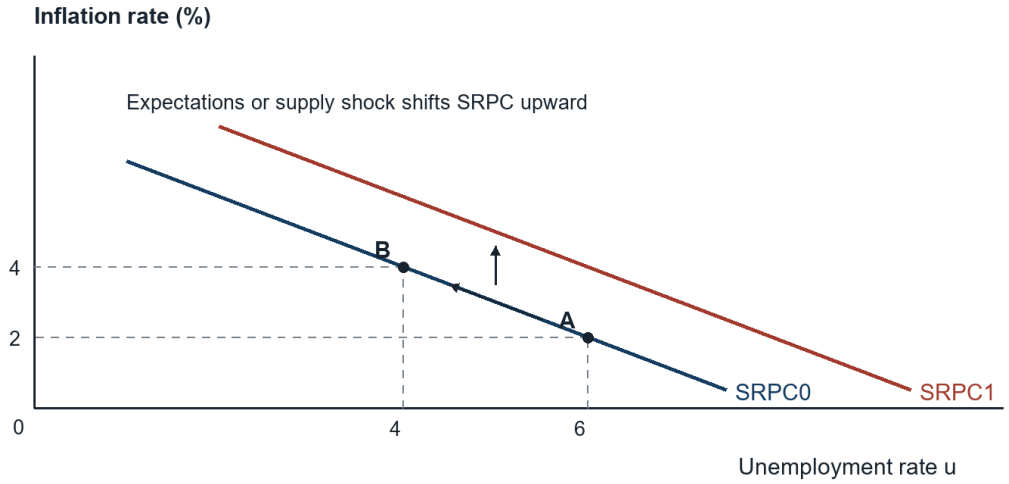
Common error. This static graph directly shows a higher price level and lower output, not the duration of inflation. In a Phillips Curve diagram, the same adverse shock shifts the short-run relationship upward; it is not a movement along an unchanged curve.

Graph 14 Short Run Phillips Curve

How to distinguish a demand-driven trade-off from a change in the inflation-unemployment relationship.

Axes and symbols. The horizontal axis measures the unemployment rate u and the vertical axis measures the inflation rate in per cent. SRPC0 is the initial short-run curve. SRPC1 is a higher curve representing changed expectations or an adverse supply shock.

Why the curves have these shapes. SRPC0 slopes downward when expected inflation and supply conditions are held constant. Stronger demand raises output and hiring, reducing unemployment while increasing wage and price pressure. This is a conditional short-run relationship, not a claim that unemployment mechanically determines inflation by itself.



Reading the equilibrium. A has unemployment of 6 per cent and inflation of 2 per cent. B has unemployment of 4 per cent and inflation of 4 per cent. Both points lie on SRPC0. The arrow from A to B therefore represents a movement along the same curve.

Adjustment and interpretation. A demand expansion can produce A to B. Demand contraction moves in the opposite direction. By contrast, higher expected inflation or an adverse supply shock shifts the entire curve upward to SRPC1: at any given unemployment rate, inflation is higher. A favourable supply development or lower expectations can shift it downward.

Illustration. If firms receive more orders, they may hire workers and raise prices, producing the A-to-B combination. If imported fuel costs instead rise sharply, inflation and unemployment can both worsen as the short-run curve shifts; that is not the A-to-B movement.

Exam conclusion. The short-run Phillips Curve shows a trade-off only while expectations and supply conditions are held constant. Identify the cause before deciding whether to draw a movement or a shift.

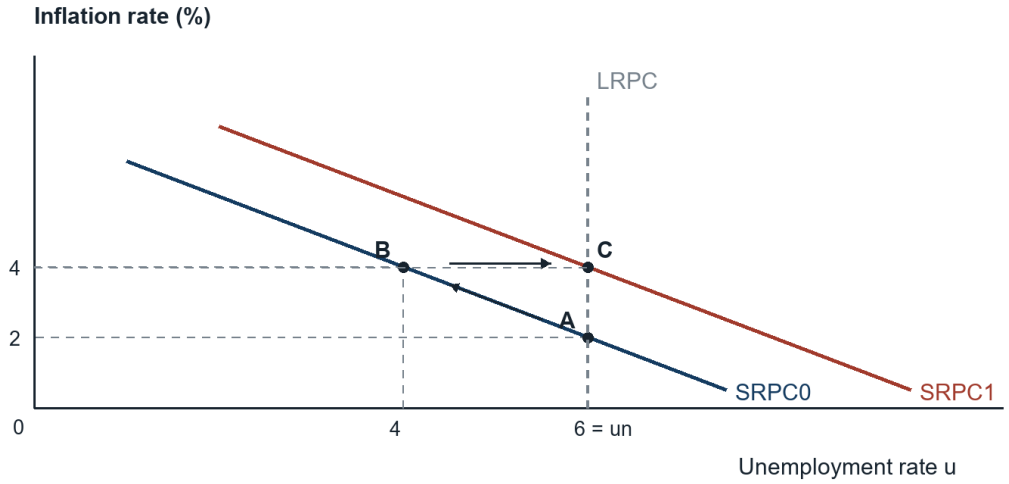
Common error. Never label the vertical axis as the price level. The modern graph uses the inflation rate; Phillips's original study concerned wage inflation rather than the price-inflation version used here.

Graph 15 Long Run Phillips Curve and Expectations

Why an initial fall in unemployment need not last after inflation expectations adjust.

Axes and symbols. Unemployment u is on the horizontal axis and inflation on the vertical axis. LRPC is vertical at the natural unemployment rate un, illustrated as 6 per cent. SRPC0 and SRPC1 correspond to lower and higher expected inflation.

Why the curves have these shapes. Each short-run curve slopes downward under fixed expectations. The long-run curve is vertical because once actual and expected inflation agree, the standard model returns unemployment to its natural rate. This rate includes frictional and structural unemployment and is not zero.



Reading the equilibrium. A lies on SRPC0 and LRPC: unemployment is 6 per cent and inflation is 2 per cent. B lies on SRPC0 at 4 per cent unemployment and 4 per cent inflation. C lies on SRPC1 and LRPC at 6 per cent unemployment and 4 per cent inflation.

Adjustment and interpretation. An unanticipated demand expansion first moves the economy from A to B. Workers and firms then revise inflation expectations and adjust wage and price decisions. SRPC shifts upward to SRPC1, and unemployment returns to un at C. Comparing A with C reveals the long-run outcome: higher inflation but no lasting unemployment reduction.

Illustration. Initially, people may expect 2 per cent inflation. If actual inflation rises to 4 per cent and this rate becomes expected, the initial employment advantage disappears in the model. A continuing 4 per cent inflation rate cannot permanently keep unemployment at 4 per cent.

Exam conclusion. There is no permanent long-run trade-off between inflation and unemployment in this expectations-augmented model. Keeping unemployment below un through repeated surprises would require accelerating inflation under adaptive expectations.

Common error. Do not draw B to C as movement along SRPC0. Expectations change the curve itself. Structural changes can alter un, so the position of LRPC is not fixed forever.

Comparing the Graphs

Graph family	Horizontal axis	Vertical axis
Labour market	Employment N	Real wage W/P
Production function	Employment N	Real output Y
AD–AS	Real output Y	Price level P
Saving and investment	Planned S and I	Interest rate r
Effective demand	Employment N	Expected or required proceeds
Expenditure cross	Real income Y	Planned expenditure AE
Phillips Curve	Unemployment rate u	Inflation rate

Initial change	Price level	Output	Interpretation
AD right with upward SRAS	Higher	Higher in short run	Demand-pull pressure
SRAS left with unchanged AD	Higher	Lower	Cost-push pressure
AD left with upward SRAS	Lower	Lower	Demand-led deflation
AD left with horizontal SRAS	Unchanged	Lower	Fixed-price Keynesian case
AD right with vertical LRAS	Higher	Unchanged	Long-run classical case

Time matters. These AD–AS comparisons describe price levels, not directly inflation rates. A sequence of price-level increases is required to establish continuing inflation.

Movement or shift. A change in an axis variable generally means movement along a given curve. A change in an underlying determinant shifts the curve. State what is held constant before tracing the result.

Equilibrium is not a welfare judgement. Equilibrium is not always full employment. In the effective-demand graph, Ne can be below Nf; in the expenditure cross, Y = AE does not identify potential output by itself.

How to Draw and Explain a Graph in the Exam

1 Select the correct model. Use the labour market for classical real-wage adjustment; D and Z for Keynesian effective demand; AD–AS for inflation and output; Phillips curves for inflation and unemployment.

2 Label the axes before drawing. Write both the variable and its meaning. Distinguish employment from unemployment, proceeds from prices, and the price level from the inflation rate.

3 Draw the initial curves. Use the slope required by the model and name each curve near a clear endpoint. Reserve space for a second curve if a shift is needed.

4 Mark the original position. Label E0 or E and use dashed projections to the axes. A simple qualitative graph normally needs named values, not an invented numerical scale.

5 State the trigger and show the change. For example, an oil-cost increase shifts SRAS left. Label the new curve and E1. Use a clear arrow to show the direction of the shift.

6 Compare outcomes explicitly. Write a connected sentence: the price level rises from P0 to P1 while output falls from Y0 to Y1. Explain why the trigger produces those changes.

7 Add one example and one qualification. An example links the graph to economic behaviour. A qualification states an assumption, such as unchanged expectations or fixed productive capacity, rather than claiming the result holds in every situation.

A reusable explanation pattern

The horizontal axis shows … and the vertical axis shows …. Initially, the economy is at …. A change in … shifts … from … to …. The new intersection is …. Therefore, … rises or falls because …. This conclusion assumes ….

Source and scope guide

Primary scope: the uploaded BAHUM107 Macro Economics syllabus, version 1.0, Module 2, Determination of Output, Employment and Inflation, PDF page 1. The syllabus specifies the topics, not a mandatory diagram list.

Companion reference: BAHUM107 Module 2 Notes and Exam Answers, notes pages 3–14, created earlier in this project. Its original figure explanations and source guide inform the diagrams expanded here. The uploaded introductory slide decks supply background rather than a complete Module 2 graph set.

Supporting conceptual references carried forward from the original guide: Keynes, The General Theory, Chapter 3 on effective demand and Chapter 19 on money wages; OpenStax, Principles of Macroeconomics 3e, sections 11.3–11.6 and 12.1–12.3, plus its expenditure-output model appendix. See the original guide's Sources and Further Reading page for source links.

The production function, classical long-run AD–AS diagram and expenditure cross are supporting teaching diagrams. The purchasing-power graph illustrates one inflation effect; there is no single standard graph that represents every social and resource cost of inflation.